

EXHALE: Early-Phase Update Log

(AIOLOS port, trace metals, metal cooling)

*This document collects, in full, the early-phase physics / implementation memos of EXHALE that predate the main Update_EXHALE phase log: the AIOLOS feature port (opacity-model dispatcher and trace metals), the metal-cooling implementation, and the Phase-2 metal extensions. Each **Part** below is one original memo, kept verbatim. Related documents: the $H\alpha/H\beta$ transmission physics is in its own note transmission_spectrum; the solver/numerics history is in Update_EXHALE_solver; the ongoing phase work and version history are in Update_EXHALE; the practical guide is EXHALE_user_manual.*

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Part I

AIOLOS→ATES feature port: opacity dispatcher and trace metals (2026-04-29)

Context note (2026). This memo was originally written for the ATES_extended fork (decoupled coronal metal solver) and has been copied into EXHALE, the main development version. In EXHALE the metal physics lives in the coupled MINPACK ionization system (Badnell RR+DR recombination, nitrogen, charge transfer with H, an A/C/P/T opacity-model dispatcher, and runtime metals.inp/opacity.inp); see cooling_formulas for the closed-form metal line-

*cooling formulas EXHALE now uses, and Update_EXHALE §7 for the Phase-2 metal line-cooling validation gate.
(Updated 2026-08-15: the note originally pointed at ATES_metal_cooling.tex, which no longer exists.)*

1. Overview

This memo documents the first round of porting selected AIOLOS (Schulik & Booth 2023) features into ATES v2.0 (Caldirola+ 2021, 2022). Both codes are exoplanet atmospheric photoionization-hydrodynamics codes but with complementary strengths. ATES has a polished H/He photoionization solver (MINPACK nonlinear equilibrium), a Tk-based input GUI, and a streamlined Fortran code base; AIOLOS supports arbitrary numbers of species, a general photochemistry network, eight opacity models, and multi-band radiative transfer. The chosen porting strategy is “keep ATES as the base and incrementally bring AIOLOS features over.”

The full plan is split into three phases:

- **Phase 1** — opacity model dispatcher and multi-band infrastructure (smallest scope; preserves existing ATES behavior).
- **Phase 2** — extend the ionization network beyond H/He by adding trace metal species (C, N, O).
- **Phase 3** — multi-fluid hydrodynamics (conservative state vector $3 \rightarrow 3N$; out of scope for this round, intentionally deferred to a separate sub-project).

This memo records the completion status of Phase 1 and Phase 2 along with intentional simplifications and known limitations. All edits are isolated to a fresh directory ATES/ATES_extended/; the upstream EXHALE and the local ATES_v1_my fork are untouched.

2. Working directory layout

EXHALE was copied verbatim into ATES_extended/ and the following files were added or modified:

ATES_extended/

run_EXHALE.sh (modified: compile order)

src/

modules/

init/

parameters.f90 (modified: new globals)

set_energy_vectors.f90 (modified: dispatcher call)

functions/

cross_sec_metals.f90 (new: Verner+1996)

radiation/

opacity_models.f90 (new: A/C/P/T dispatcher)

metals.f90 (new: metals registry)

```

    metals_solve.f90                (new: coronal balance helpers)
    metals_drive.f90                (new: cell-by-cell driver)
files_IO/
    opacity_input_read.f90          (new: opacity.inp parser)
    write_metals_output.f90         (new: output writer)
post_process/
    post_process_adv.f90            (modified: 9 lines added)
inputdata/
    HI_sample.opa                  (new: sample table)
    opacity.inp.example            (new)
    metals.inp.example             (new)
    README.opacity                 (new)
docs/
    my_memo.cls                    (copy)
    aiolos_port_memo.tex           (this document)

```

The build system is the ATES-original explicit compile-order list in `run_EXHALE.sh`; new modules were inserted at positions that satisfy their module-use dependencies.

3. Phase 1 — Opacity model dispatcher

3.1 Motivation

The original ATES hard-codes its H/He photoionization cross sections as analytic functions in `cross_sec.f90`: σ_{HI} , σ_{HeI} , σ_{HeII} , and $\sigma_{\text{HeI}(2^3\text{S})}$. At startup, `set_energy_vectors.f90` prebakes these into the global vectors $s_{\text{HI}}, s_{\text{HeI}}, s_{\text{HeII}}, s_{\text{HeI,TR}} \in \mathbb{R}^{N_t}$, which are then consumed by `util_ion_eq.f90`, `Cool_coeff.f90`, and downstream modules.

AIOLOS’ `opacities.cpp` dispatches across eight models (U, C, P, F, M, D, T, K) inside `c_Species::update_opacities()`. Of these, the four that are physically compatible with ATES’ scope (EUV-driven photoionization, no thermal IR transfer) — A (analytic), C (constant), P (Robinson & Catling pressure broadening), T (tabulated) — are ported. The F/M/D/K models presuppose thermal IR Planck/Rosseland opacities and would require adding a new radiative-transfer solver to ATES first.

3.2 Dispatcher design

A new module `opacity_models.f90` exposes the dispatcher entry point:

```

double precision function photoion_sigma(sp, E)
    character(len=*) , intent(in) :: sp      ! 'HI', 'HeI', 'HeII', 'HeITR'
    real*8,           intent(in) :: E        ! [eV]

```

! returns sigma in [1e-18 cm²] (ATES internal unit)

Branch logic:

$$\sigma(\text{sp}, E) = \begin{cases} \sigma_{\text{analytic}}(\text{sp}, E) & \text{model} = \text{A (default)} \\ f_{\text{sp}} \sigma_{\text{analytic}}(\text{sp}, E_{\text{th}}^{\text{sp}}) \cdot \mathbf{1}[E \geq E_{\text{th}}^{\text{sp}}] & \text{model} = \text{C} \\ \text{(same as C; broadening factor applied separately)} & \text{model} = \text{P} \\ \text{loglog interpolation on tab_sp} & \text{model} = \text{T} \end{cases}$$

The “analytic” branch simply forwards to the existing ATES functions, so under the default model A the dispatcher is a transparent wrapper and the prebaked vectors s_{HI} , etc. remain bit-identical to the original. The pressure-broadening factor used by model P,

$$f_{\text{pres}}(p) = 1 + a \left(\frac{p}{p_{\text{pivot}}} \right)^n, \quad p_{\text{pivot}} = 10^5 \text{ dyne cm}^{-2} = 0.1 \text{ bar},$$

is exposed as a separate function `opacity_pT_factor(p)`, mirroring the AIOLOS implementation (`opacities.cpp`, Robinson & Catling 2013, *Nat. Geo.*). It returns 1 in any model other than P. The cell-level multiplication is currently a no-op because Phase 1 only replaces the prebaked global vector; turning it on per cell is a Phase-1d extension (see limitations).

3.3 Input file: `opacity.inp`

The existing `input.inp` parser (`input_read.f90`) is position-based and brittle. To avoid regressions, a separate optional file `opacity.inp` was introduced. The format is one KEY = VALUE per line; # starts a comment; blank lines are ignored:

```
OPACITY_MODEL      = T           # A | C | P | T (default A)
OPA_CONST_HI       = 1.0         # multiplier for C/P models
OPA_CONST_HEI      = 1.0
OPA_CONST_HEII     = 1.0
OPA_CONST_HEITR    = 1.0
OPA_PB_FACTOR      = 0.0         # Robinson & Catling a
OPA_PB_EXPONENT    = 1.0         # n
OPA_PB_PIVOT       = 1.0e5       # dyne/cm^2
OPA_FILE_HI        = inputdata/HI_sample.opa
OPA_FILE_HEI       = inputdata/HeI.opa
OPA_FILE_HEII      = inputdata/HeII.opa
OPA_FILE_HEITR     = inputdata/HeITR.opa
```

If the file is absent, every key keeps its default from `parameters.f90` (model A, factor 1.0, no table paths) and there is no behavior change. Unrecognized keys produce a warning and are skipped.

3.4 .opa table format

Two columns, whitespace separated, no header:

column 1: photon energy [eV], strictly increasing
column 2: photoionization cross section [10^{-18} cm^2]

The cross-section unit follows the ATES internal convention (`cross_sec.f90` returns values in 10^{-18} cm^2). Interpolation is log-log linear; out-of-range queries clamp to the nearest endpoint. When either endpoint of the bracket is non-positive, the routine falls back to linear-in-y interpolation. This format is semantically equivalent to AIOLOS' *.opa files but is given a distinct extension to make the unit convention explicit.

The shipped `inputdata/HI_sample.opa` is a 13-point evaluation of the analytic HI curve (at 13.61, 14, 16, 20, 25, 35, 50, 75, 100, 200, 500, 1000, 5000 eV). Running model T with this file should reproduce the model A result to within interpolation accuracy — a useful regression check.

3.5 Call flow

The cross-section assignment in `set_energy_vectors.f90` now reads:

```
! AIOLOS port: read optional opacity.inp + load tables
call read_opacity_input
call load_opacity_tables

! original: s_hi = (/ (sigma(e_v(i),ih), i = 1,Nl) /)
! new: route through dispatcher
s_hi    = (/ (photoion_sigma('HI',   e_v(i)), i = 1,Nl) /)
s_hei   = (/ (photoion_sigma('HeI',  e_v(i)), i = 1,Nl) /)
s_heii  = (/ (photoion_sigma('HeII', e_v(i)), i = 1,Nl) /)
s_heiTR = (/ (photoion_sigma('HeITR', e_v(i)), i = 1,Nl) /)
```

No other ATES module cares where the cross-section vectors come from, so `Cool_coeff.f90`, `ionization_equilibrium.f90`, `util_ion_eq.f90`, `post_process_adv.f90`, etc., remain unchanged.

4. Phase 2 — Trace metals (C, N, O)

4.1 Motivation and coupling-strength decision

The original ATES treats only H + He. Metal photoionization signatures (C, N, O, ...) are increasingly important observationally for exoplanet atmospheres (e.g., HD 209458b OI escape, Vidal-Madjar+ 2003), so adding them is a natural next step. There are two architectural options for adding metals, distinguished by how strongly metals are coupled to the H/He dynamics:

1. **Strong coupling:** integrate metals into the H/He MINPACK nonlinear ionization-equilibrium system. This grows N_{eq} (e.g. $4 \rightarrow 8$ with C/N/O), forces every nonlinear-solver call to reshape, and requires adding metal cross-section vectors, initial-guess logic, output columns, and metal line cooling for self-consistent thermal balance. Touches almost every existing file.
2. **Weak coupling (trace approximation):** solve the metals *after* H/He has converged. Given a cell's (n_e, T, J_ν) from the H/He solution, close the metal balance with the coronal-equilibrium relation

$$\Gamma_k(j) n_X^k(j) = \alpha_{\text{rec}}(k+1 \rightarrow k, T_j) n_e(j) n_X^{k+1}(j).$$

The H/He system is left untouched.

This work adopts option (b). Reasons:

- Non-invasive on the ATES core (MINPACK system, Jacobian, input parser all remain unchanged).
- Trace metals contribute negligibly to the thermal/ionization balance at solar-like abundances ($n_X/n_H \sim 10^{-4}$), so the approximation is consistent with itself.
- Phase 2 can be reviewed and tested as a self-contained increment; strong coupling can become a separate fork later.
- Migration to strong coupling later can reuse the cross sections, recombination tables, and abundance registry built here.

4.2 Photoionization cross sections: Verner+1996 fit

The new module `cross_sec_metals.f90` implements the Verner et al. (1996, ApJ 465, 487) analytic fit:

$$\sigma(E) = \sigma_0 \cdot F(y), \quad y = E/E_0,$$

$$F(y) = [(y-1)^2 + y_w^2] y^{-Q} (1 + \sqrt{y/y_a})^{-P}, \quad Q = 5.5 + l - \frac{1}{2}P.$$

The seven parameters ($E_{\text{th}}, E_0, \sigma_0, y_a, P, y_w, l$) are hard-wired for six ions (CI, CII, NI, NII, OI, OII), valence shell only, using the published Table 1 values. σ_0 is given in megabarns ($1 \text{ Mb} = 10^{-18} \text{ cm}^2$), so the result is already in ATES' internal unit and no conversion is needed.

Sanity check: evaluating the same formula with HI parameters gives $\sigma_{\text{HI}}(13.6 \text{ eV}) = 6.346 \text{ Mb}$, in agreement with the standard value 6.3 Mb to 1%. This validates both the formula and the implementation.

Adding more ions (CIII, NIII, OIII, Mg, Si, Fe, ...) is a single-row extension in the lookup select case block of `cross_sec_metals.f90`.

4.3 Recombination coefficients: A&P 1973 RR + SVS 1982 DR

`metals_solve.f90` computes total recombination as the sum of radiative (RR) and dielectronic (DR) channels:

$$\alpha_{\text{rec}}(T) = \alpha_{\text{RR}}(T) + \alpha_{\text{DR}}(T).$$

RR uses the Aldrovandi & Pequignot (1973, A&A 25, 137) single power-law form:

$$\alpha_{\text{RR}}(T) = A_{\text{RR}} \left(\frac{T}{10^4 \text{ K}} \right)^{-\eta_{\text{RR}}} [\text{cm}^3 \text{ s}^{-1}].$$

DR uses the Shull & Van Steenberg (1982, ApJS 48, 95) Burgess-style 4-parameter form:

$$\alpha_{\text{DR}}(T) = A_{\text{DR}} T^{-3/2} \exp(-T_0/T) [1 + B_{\text{DR}} \exp(-T_1/T)].$$

Both are keyed by the *daughter* ion (e.g. $\alpha_{\text{RR}}(\text{OI}, T)$ is the rate for $\text{OII} + e^- \rightarrow \text{OI}$). Built-in coefficients for the six ions matching our Verner+1996 cross-section table:

Daughter	A_{RR}	η_{RR}	A_{DR}	B_{DR}	T_0 [K]	T_1 [K]
CI	4.7×10^{-13}	0.624	2.54×10^{-4}	4.21×10^{-2}	1.57×10^5	3.74×10^5
CH	2.3×10^{-12}	0.645	6.15×10^{-3}	5.88×10^{-2}	1.41×10^5	1.41×10^5
NI	4.1×10^{-13}	0.608	1.50×10^{-4}	0.0	2.99×10^5	0.0
NII	2.2×10^{-12}	0.639	8.46×10^{-2}	0.420	2.83×10^5	4.51×10^5
OI	3.1×10^{-13}	0.678	1.11×10^{-3}	9.25×10^{-2}	1.75×10^5	1.45×10^5
OII	2.0×10^{-12}	0.646	5.07×10^{-3}	0.181	1.98×10^5	3.35×10^5

A_{RR} in $\text{cm}^3 \text{ s}^{-1}$; A_{DR} in $\text{cm}^3 \text{ K}^{3/2} \text{ s}^{-1}$.

The module exposes three accessors — `alpha_rr_metal`, `alpha_dr_metal`, and `alpha_rec_metal` — so that RR and DR contributions can be inspected independently. The coronal solver (`coronal_ratio`) consumes only the total `alpha_rec_metal`.

Why A&P + SVS rather than V&F 1996 + Badnell? The motivation to add DR was the original concern that DR can be non-negligible at $T \gtrsim 2\text{--}3 \times 10^4 \text{ K}$. Shull & Van Steenberg’s published fits cover exactly the six daughters we need, with peer-reviewed parameters. The more modern V&F 1996 RR + Badnell (2003+) DR series would be a quantitatively better choice (V&F is $\sim 5\times$ larger than A&P for CI at $T = 10^4 \text{ K}$, for instance), but the V&F parameters for several ions of interest were not available with high enough confidence to commit to literal values here. The architecture is deliberately designed so that swapping in V&F or Badnell coefficients is a parameter-block edit only — no caller change required.

Behavior across temperature. At $T = 10^4 \text{ K}$ (the ATES escape-region working point) DR is negligible for all six channels, so the total matches A&P alone. DR becomes comparable to RR at $T \sim 3 \times 10^4 \text{ K}$ (e.g. $\text{DR/RR} \approx 1.1$ for CI, 4.3 for OI), and dominates by $T \sim 10^5 \text{ K}$ ($\text{DR/RR} \approx 15$ for CI, 96 for OI). The upgrade therefore extends the formula’s validity range upward without changing predictions in the regime where ATES typically operates.

4.4 Coronal-equilibrium solve

For each enabled element X , only two stages (neutral and first ion) are tracked. Steady state and element conservation give

$$\Gamma_{\text{neutral}} n_X^0 = \alpha_{\text{rec}}(\text{next}, T) n_e n_X^+, \quad n_X^0 + n_X^+ = n_X^{\text{total}} = (n_X/n_H) n_H(j).$$

Defining $r \equiv n_X^+ / n_X^0$:

$$r = \frac{\Gamma_{\text{neutral}}}{\alpha_{\text{rec}} n_e}, \quad n_X^0 = \frac{n_X^{\text{total}}}{1+r}, \quad n_X^+ = n_X^{\text{total}} \frac{r}{1+r}.$$

The photoionization rate Γ is integrated over the local attenuated flux $F_{\text{loc}}(E)$:

$$\Gamma_{\text{neutral}}(j) = \sum_{k=1}^{N_l} \Delta E_k \frac{F_{\text{loc}}(j, E_k)}{E_k} \sigma(E_k) 10^{-18}$$

(ΔE_k is `de_v(k)`; F_{loc} in $\text{erg cm}^{-2} \text{s}^{-1} \text{eV}^{-1}$; σ in 10^{-18}cm^2 ; the explicit 10^{-18} converts σ to physical cm^2).

F_{loc} is rebuilt with the same expression that `PH_heat_HHe` (`util_ion_eq.f90`) uses internally, from the H/He column densities and prebaked cross-section vectors:

$$\tau(E_k, j) = 10^{-18} [s_{\text{HI}}(k) N_1(j) + s_{\text{HeI}}(k) N_{1.5}(j) + s_{\text{HeII}}(k) N_2(j) + (\text{HeITR term})]$$

$$F_{\text{loc}}(j, E_k) = \frac{F_{\text{XUV}}(k) e^{-\tau(E_k, j)}}{1 + a_\tau \tau(E_k, j)}$$

(a_τ is the ATES “2D approximate method” factor.) This block is recomputed inside `solve_metals_post` (`metals_drive.f90`) in ~ 10 lines. The duplication is intentional — preserving ATES core code uncorrected was the higher priority.

4.5 Module structure

File	Responsibility
<code>cross_sec_metals.f90</code>	Verner+1996 cross-section function and built-in fit table
<code>metals.f90</code>	Active-metals registry, <code>metals.inp</code> parser, prebake cross sections on the energy grid, Γ integral
<code>metals_solve.f90</code>	A&P 1973 RR + SVS 1982 DR lookup; total $\alpha_{\text{rec}} = \alpha_{\text{RR}} + \alpha_{\text{DR}}$; $r = \Gamma / (\alpha_{\text{rec}} n_e)$ helper
<code>metals_drive.f90</code>	Attenuated-flux recomputation cell by cell, calls <code>coronal_ratio</code> for every active metal, fills neutral/ion density arrays
<code>write_metals_output.f90</code>	Writer for output/ <code>Metals_ioniz_adv.txt</code>

Layered dependencies: `cross_sec_metals` $\leftarrow$ `metals` $\leftarrow$ `metals_solve`, `metals_drive` $\leftarrow$ `write_metals_output`. Every entry point checks `n_metals == 0` and returns immediately when metals are disabled, so the inactive case is a true no-op.

4.6 Input file: `metals.inp`

```
# metals.inp -- one ion per line: ION abundance(rel. H by number)
# Recognized ions: CI, NI, OI, CII, NII, OII
# Asplund+2009 solar abundances:
CI      2.69e-4
NI      6.76e-5
OI      4.90e-4
```

Convention: the user lists *neutrals* (CI, NI, OI), and the abundance is the *element* number ratio n_X/n_H (the sum across all stages). The solver pushes both the listed neutral and its first ion (e.g. CI $\rightarrow$ CII) and closes the two-stage balance. If `metals.inp` is absent, `n_metals = 0` and every downstream code path is a no-op.

4.7 Output file: `Metals_ioniz_adv.txt`

Written once per post-processing cycle. Format:

```
# r    n(CI)    n(CII)    n(NI)    n(NII)    n(OI)    n(OII)
1.0000e+00  3.21e+04    ...
1.0050e+00  3.18e+04    ...
...
```

Columns appear in the order metals were registered in `metals.inp`; the first line is a comment header naming them. r is the dimensionless radius from `global_parameters.f90`. Densities are in cm^{-3} .

4.8 ATES-core integration footprint

A single existing file, `post_process_adv.f90`, was modified, and the change is minimal:

- use `metals`, use `metals_drive`, use `metals_output` (3 lines)
- Local buffers `n_metal_neutral(max_metals, 1-Ng:N+Ng)` and `n_metal_ion(. . .)` (1 line)
- Right after the existing call `write_output(...)` at the end of the post-processing loop:

```
if (n_metals > 0) then
  call solve_metals_post(nhi, nhii, nhei, nheii, nheiTR, T_K, &
                        n_metal_neutral, n_metal_ion)
  call write_metals_output(n_metal_neutral, n_metal_ion, 'ad')
endif
```

These nine lines are the *entire* ATES-core change for the ionization side. (Phase 2f below adds a one-line modification to `T_equation.f90` and a few more to `post_process_adv.f90` when metal cooling is enabled.)

4.9 Metal line cooling (Phase 2f)

The trace approximation makes one prediction that, if accurate, would be uncomfortable: in partially ionized layers OI $63\ \mu\text{m}$ can dominate H Ly α as a coolant by orders of magnitude. To make ATES capable of testing this self-consistently, we port the closed-form cooling coefficients from AIOLOS' `photochem.cpp` (Black 1981 fits):

$$\Lambda_X(T_e) = a_X + b_X e^{-T_X/T_e} \left(1 + (T_e/T_{X,s})^{p_X}\right) \quad [\text{erg cm}^3 \text{ s}^{-1}],$$

with (a, b, T, T_s, p) tabulated below for $X \in \{\text{CI}, \text{CII}, \text{OI}, \text{OII}\}$. NI and NII are set to zero (no AIOLOS reference; user can plug in Hollenbach & McKee 1989). Cooling rate per cm^3 is $n_X n_e \Lambda_X(T_e)$.

X	a	b	T_X [K]	$T_{X,s}$ [K]	p_X
CI	1.0×10^{-24}	3.1×10^{-20}	15162	2.0×10^4	1.5
CII	1.5×10^{-23}	3.1×10^{-20}	45162	0.75×10^4	1.5
OI	5.5×10^{-24}	1.1×10^{-20}	30162	0.75×10^4	0.5
OII	0	5.1×10^{-20}	35162	0.75×10^4	0.5

Coupling into the implicit T solve

This is the first part of the port that creates a real two-way coupling between metals and the H/He thermal balance. ATEs' T_equation already evaluates $\Lambda_{H/He}(T)$ inside each MINPACK iteration to maintain implicit stability; the same pattern is reused. The addition is:

```
! T_equation.f90 (one-line addition to total cooling)
m_cool = params(13)                                ! NEW
cool    = ne*(brem + coex + reco + coio)/q0 + m_cool ! MODIFIED
```

The metal cooling enters as a number precomputed for each cell (params(13)) rather than being recomputed at every MINPACK iteration. This is a deliberate one-iteration lag: post_process_adv runs an outer convergence loop do k = 1, 10, and each pass re-solves the metal coronal balance (solve_metals_post) and re-evaluates Λ_X at the current T_K , so at convergence T and $\{n_X^k\}$ are mutually consistent. Within one MINPACK call the metal cooling is a constant, which avoids re-doing the cell loop for every Newton iteration ($\sim 10\times$ less work) at the cost of making the outer fixed-point iteration responsible for closing the coupling. The convergence is fast in practice because $\Lambda_X(T)$ is smooth near $T \sim 10^4$ K.

The post_process_adv edit moves the metal solve from “after the loop” (Phase 2e) to “inside the loop” (Phase 2f) and adds the cooling evaluation:

```
do k = 1, 10
  ...
  call PH_heat_HHe(...) ! existing: heating
  theat = theat / q0
  if (n_metals > 0) then ! NEW (Phase 2f)
    call solve_metals_post(..., n_metal_neutral, n_metal_ion)
    call eval_metal_cooling(T_K, n_metal_neutral, n_metal_ion, &
                           ne, metal_cool_per_cell)
  endif
  do j = ...
    paramsT(13) = metal_cool_per_cell(j) / q0 ! NEW
    call hybrd1(T_equation, ..., paramsT)
  enddo
  ...
enddo
```

Where the metal cooling matters

A direct evaluation of $\Lambda_X(T)$ vs the H Ly α coefficient $\Lambda_{\text{Ly}\alpha}(T) = 7.5 \times 10^{-19} e^{-118400/T}$ shows the regime split. At $T = 5 \times 10^3$ K, $\Lambda_{\text{OI}} \approx 5 \times 10^{-23}$ vs $\Lambda_{\text{Ly}\alpha} \approx 6 \times 10^{-31}$ — 8 orders of magnitude separation per ion-pair. Multiplied by solar $n_{\text{O}}/n_{\text{H}} \sim 5 \times 10^{-4}$, OI cooling dominates by $\sim 10^4$ in the partially ionized layer ($T \lesssim 8000$ K). By $T = 2 \times 10^4$ K Ly α surpasses metal cooling *per ion pair*, but by then most O is ionized so the relevant question is the OII contribution, which is also tabulated. Adding metals can therefore lower the steady-state temperature in the lower atmosphere and (indirectly) affect the mass-loss rate.

5. Verification

Compilation. All 39 modules were compiled in dependency order with gfortran 13.1.0 plus `-fopenmp -O0`, as listed in `run_EXHALE.sh`. Zero warnings, zero errors.

Default-behavior preservation. With both `opacity.inp` and `metals.inp` absent (`opacity_model = 'A'`, `n_metals = 0`), every new module either returns immediately or forwards to the existing analytic functions. The ATES result is bit-identical to the upstream code — no new code path activates a branch.

Verner unit/formula check. Evaluating `verner_F` with HI parameters yields $\sigma(13.6\text{eV}) = 6.346\text{Mb}$, matching the canonical 6.3Mb.

Recombination temperature scan. Evaluating $\alpha_{\text{rec}} = \alpha_{\text{RR}} + \alpha_{\text{DR}}$ for the six metal channels at $T \in \{5 \times 10^3, 10^4, 3 \times 10^4, 10^5\}$ K gives the expected behavior: at $T = 10^4$ K the DR contribution is $\leq 0.2\%$ for all channels, so total matches the A&P value exactly; at $T = 3 \times 10^4$ K DR becomes comparable (DR/RR ≈ 1.1 for CI, 4.3 for OI); by $T = 10^5$ K DR dominates by a factor of ~ 10 –100. This is the qualitative signature of correct DR low-temperature behavior.

Metal-cooling temperature scan. Λ_{OI} vs the H Ly α coefficient $\Lambda_{\text{Ly}\alpha} = 7.5 \times 10^{-19} e^{-118400/T}$:

- $T = 5 \times 10^3$ K: $\Lambda_{\text{OI}} \approx 5 \times 10^{-23}$, $\Lambda_{\text{Ly}\alpha} \approx 6 \times 10^{-31}$ (8 orders of magnitude separation).
- $T = 10^4$ K: $\Lambda_{\text{OI}} \approx 1.2 \times 10^{-21}$, $\Lambda_{\text{Ly}\alpha} \approx 5.4 \times 10^{-24}$ ($\sim 220\times$ separation per ion-pair).
- $T = 3 \times 10^4$ K: $\Lambda_{\text{OI}} \approx 1.2 \times 10^{-20}$, $\Lambda_{\text{Ly}\alpha} \approx 1.5 \times 10^{-20}$ (comparable; crossover region).

With solar O abundance ($n_{\text{O}}/n_{\text{H}} \sim 5 \times 10^{-4}$), metal cooling is the dominant coolant up to $T \approx 1.5 \times 10^4$ K where most O is still neutral; above this the OI density falls and Ly α takes over. This is the well-known partially-ionized layer “OI thermostat” — the qualitative signature ATES should now reproduce when metals are enabled.

Not done. Physical validation of an actual run — comparing `Metals_ioniz_adv.txt` to published profiles (e.g. Lampón+ 2020 OI/CII distributions) — requires the user to execute `run_EXHALE.sh` with appropriate input files. Compile correctness and *functional* correctness are distinct; this memo guarantees only the former.

6. Known limitations and intentional simplifications

6.1 Phase 1 (opacity)

1. **Model P is not yet applied per cell.** The pressure-broadening factor `opacity_pT_factor(p)` is exposed but `set_energy_vectors` only does a spatially uniform prebake. To multiply by the local cell pressure one must either rescale $s_X(k) \rightarrow s_X(k) \cdot f_{\text{pres}}(p_j)$ at every equilibrium call, or restructure the pipeline so that σ is no longer prebaked ($\sim 10\%$ performance impact). Models C and T work correctly.
2. **F/M/D/K models not ported.** Freedman, Malygin/Semenov, dust, and Kombined models all presuppose Planck-/Rosseland thermal IR opacities, and ATES has no thermal radiative-transfer module. Adding any of these requires first writing a separate transport solver (e.g. flux-limited diffusion).
3. **No GUI integration of `input.inp`.** The new keys live only in the separate `opacity.inp` file. The ATES Python/Tk GUI (`src/utis/EXHALE_interface_main.py`) was not modified — GUI integration is a regression-prone change, so the opt-in side-car file was chosen instead. A follow-up could add an “Advanced opacity” panel to the GUI.

6.2 Phase 2 (metals)

1. **nheiii dropped from n_e .** `solve_metals_post` does not receive `nheiii` from the caller and therefore uses $n_e \approx n_{\text{HII}} + n_{\text{HeII}}$. In the EUV-driven escape region, $n_{\text{HeIII}} \ll n_{\text{HII}}$, so this is well within the trace-metal approximation. Tightening it requires passing `nheiii` through the `post_process_adv` interface as well.
2. **Only two ionization stages (neutral + first ion).** Only X^0 and X^+ are tracked; any X^{2+} produced by photoionization of X^+ is dropped, so element conservation is violated when n_X^{2+} is non-negligible ($n_X^0 + n_X^+ < n_X^{\text{total}}$). For systems whose EUV photon distribution does not extend far above the second-ionization thresholds ($\gtrsim 25\text{--}35\text{ eV}$), the error is quantitatively acceptable. Generalizing requires (a) adding CIII/NIII/OIII rows to the Verner table, and (b) extending `coronal_ratio` to an N -stage matrix system.
3. **Metal cooling: NI/NII set to zero.** The C and O cooling functions ported from AIOLOS `photochem.cpp` cover four ions (CI, CII, OI, OII) but no nitrogen entries are provided in AIOLOS either. `lambda_X('NI', T)` and `lambda_X('NII', T)` return 0. For exoplanet atmospheres with non-trivial N abundance, plug in Hollenbach & McKee 1989 NI 122/205 μm fits into the `select` case in `metals_cool.f90` (one-line entries).
4. **Metal cooling: critical-density saturation rolled into the fit.** The Black 1981 closed-form expressions implicitly assume the gas is below the critical density of each fine-structure transition, so the cooling scales linearly with n_e . At $n_{\text{H}} \gtrsim n_{\text{cr}}$ (e.g. $n_{\text{cr, OI63}\mu\text{m}} \sim 5 \times 10^5 \text{ cm}^{-3}$ for H collisions) the LTE limit applies and the actual cooling is suppressed. ATES escape regions are typically subcritical, but the lower-atmosphere boundary cell can exceed it and the cooling will be over-predicted there. Replacing the fits with the explicit two-level form $\Lambda = h\nu A n_u (1 + n/n_{\text{cr}})^{-1}$ removes this issue.

5. **Metal cooling colliders: only n_e .** The same closed-form expressions assume e^- is the dominant collider, while in cool neutral gas OI is mostly excited by H I impacts. The cooling is therefore underestimated where the ionization fraction is small ($x_e \lesssim 0.01$). The fix is to extend the formula to $\Lambda_{\text{tot}} = (n_e \gamma_e + n_{\text{HI}} \gamma_{\text{HI}}) h\nu e^{-E/kT}$ with γ_{HI} from Draine 2011, ch. 17.
6. **Two-way coupling, not full time-integration coupling.** With Phase 2f, metals participate in the steady-state thermal balance through the post-process convergence loop. They still are not solved at each RK substep of the hydro time integration — adding them there would require call sites in `RK_rhs.f90` and `ionization_equilibrium.f90`, and would expose the cooling to stiffness during transient hydro phases. For diagnostic use cases (analyzing the steady-state metal distribution and its thermal back-reaction) the post-process integration is sufficient.
7. **Verner OI absolute value caveat.** The formula itself is verified, but $\sigma_{\text{OI}}(13.62\text{eV}) = 1.08\text{Mb}$ from the Verner+1996 fit differs by roughly a factor of three from some other sources (e.g. Reilman & Manson 1979, $\sim 2.94\text{Mb}$). This reflects the characteristics of the Verner+1996 fit (resonance smoothing); *the ported code reproduces the published Verner fit exactly*. If higher accuracy is needed the user can supply a custom cross-section table as a `.opa` file — but note the current dispatcher only handles H/He, so a metals-side dispatcher would need to be added (small generalization of `cross_sec_metals.f90`).
8. **Recombination fit choice (A&P + SVS, not V&F + Badnell).** The RR side uses A&P 1973 single-power-law rather than the more accurate Verner & Ferland 1996 4-parameter fit. For the six channels present, A&P agrees with V&F to better than $\sim 10\%$ for NI but underestimates by a factor of $\sim 4\text{--}5$ for CI and OI at $T = 10^4\text{K}$. The DR side uses SVS 1982 rather than the modern Badnell (2003+) DR series; at $T \leq 3 \times 10^4\text{K}$ the two agree to within $\sim 30\%$ for our channels, but the Badnell tables are the recommended choice for any quantitative observational comparison. Both swaps are local to the parameter blocks in `metals_solve.f90`; `coronal_ratio` and `solve_metals_post` need no change.
9. **Code duplication on the caller side.** `solve_metals_post` recomputes τ and F_{loc} independently of `PH_heat_HHe`. Factoring the shared expression into a utility function would be cleaner, but (a) the two functions have different OMP layouts, and (b) keeping the existing ATES-core modules unmodified was deliberately prioritized. Performance impact is one extra cell loop per post-processing cycle ($\sim$ tens of ms).

6.3 Phase 3 (multi-fluid) — not attempted, justification

True multi-fluid hydrodynamics is genuinely new physics: each species carries its own (ρ_X, v_X, p_X) . ATES uses a single mass-momentum- energy state vector $\mathbf{u}(j) \in \mathbb{R}^3$, so converting requires:

- $\mathbf{u}(j) \in \mathbb{R}^{3N}$.
- Full rewrite of `src/modules/states/` (Apply_BC, PLM_rec, Reconstruction, Source).
- Full rewrite or dispatch for each species in `src/modules/flux/` (HLLC, ROE, Num_Fluxes).

- RHS for each species in `src/modules/time_step/RK_rhs.f90`.
- A friction (drag) source-coupling module from AIOLOS' `source.cpp` (ATES has none — need both an analytic implicit and a numeric implicit solver).
- Output format overhaul.

This touches almost every file in the ATES core and the numerical/physical verification effort is at least an order of magnitude larger than Phase 1+2. A separate “ATES-MS” fork is the safer container. The infrastructure built in Phase 1+2 (cross sections, abundance registry, opacity dispatcher) remains directly reusable in Phase 3.

7. Recommended next steps

In rough priority order:

1. **Regression test:** compare an existing EXHALE run with the equivalent ATES_extended run (no input override files). All columns in `Hydro_ioniz.txt` must be bit-identical. Any discrepancy points to a regression in the `photoion_sigma 'A'` branch or the `set_energy_vectors` edits.
2. **Model T regression test:** with `HI_sample.opa` and `OPACITY_MODEL = T`, `OPA_FILE_HI = inputdata/HI_sample.opa` (other species fall back to the analytic curve). `Hydro_ioniz.txt` should agree with the model-A run to $\lesssim 1\%$ (interpolation accuracy).
3. **Metals smoke test:** `metals.inp` containing one line `OI 4.9e-4`. After a steady-state run, check that the `OI/OII` ratio in `Metals_ioniz_adv.txt` is stable to within ~ 1 order of magnitude in the escape region and that `OI` dominates near the planetary surface ($r \sim 1$).
4. **Python plot helper:** extend `EXHALE_plots.py` to overlay `Metals_ioniz_adv.txt` on the same coordinate as `Hydro_ioniz_adv.txt`.
5. **Extend ionization stages:** add `CIII/NIII/OIII` rows to the Verner table; generalize `coronal_ratio` to a three-element system.
6. **Upgrade recombination tables:** replace the A&P 1973 RR parameter blocks in `metals_solve.f90` with V&F 1996 4-parameter values (largest improvement: $\sim 5\times$ for `CI`, `OI`), and replace SVS 1982 DR with the Badnell (2003+) DR series. Both are local edits to parameter blocks — the RR/DR formula structure and all callers are already in place.
7. **Time-loop integration:** call the metals solver from `ionization_equilibrium.f90` every equilibrium step rather than only in post-processing.
8. **Validate metal cooling effect:** compare the steady-state $T(r)$ profile from `Hydro_ioniz_adv.txt` with and without `metals.inp` present, especially in the partially ionized layer ($r \sim 1$). A drop of $\Delta T \sim 1000\text{--}3000\text{K}$ in the lower atmosphere when `OI` is enabled is the expected qualitative signature of working `OI 63  $\mu\text{m}$` cooling.

9. **Improve cooling fidelity:** add the n_{HI} collider channel and the critical-density saturation factor (see Phase 2f limitations) to `lambda_X`. Plug in NI/NII fits if N matters in the target system.
10. **Phase 3** as a separate fork for multi-fluid hydrodynamics.

Appendix A: Full file inventory

A.1 New files (12)

Path	Role	Lines
<code>src/modules/radiation/opacity_models.f90</code>	A/C/P/T dispatcher	232
<code>src/modules/files_IO/opacity_input_read.f90</code>	<code>opacity.inp</code> parser	116
<code>src/modules/functions/cross_sec_metals.f90</code>	Verner+1996 fit	121
<code>src/modules/radiation/metals.f90</code>	metals registry	162
<code>src/modules/radiation/metals_solve.f90</code>	A&P RR + SVS DR, ratio helpers	209
<code>src/modules/radiation/metals_drive.f90</code>	cell-by-cell coronal driver	92
<code>src/modules/radiation/metals_cool.f90</code>	Black 1981 cooling fits	102
<code>src/modules/files_IO/write_metals_output.f90</code>	output writer	62
<code>inputdata/HI_sample.opa</code>	13-row sample table	13
<code>inputdata/opacity.inp.example</code>	key documentation	22
<code>inputdata/metals.inp.example</code>	usage example	10
<code>inputdata/README.opacity</code>	format documentation	31

A.2 Modified existing files (5)

Path	Change
<code>src/modules/init/parameters.f90</code>	Added opacity globals and metals slots (defaults preserve original behavior)
<code>src/modules/init/set_energy_vectors.f90</code>	Replaced direct analytic calls with <code>photoion_sigma</code> ; added metals init calls
<code>src/modules/post_process/post_process_adv.f90</code>	Phase 2e: 9 lines for metals solver and writer. Phase 2f: extra metal solve + cooling evaluation inside the convergence loop, <code>paramsT(13)</code> per cell, <code>paramsT</code> grown from 12 to 13.
<code>src/modules/nonlinear_system_solver/T_equation.f90</code>	Phase 2f: read <code>params(13)</code> as metal cooling and add to total cool. Two-line addition.
<code>run_EXHALE.sh</code>	New modules added to compile order in dependency-correct positions

Appendix B: Build and run summary

Build: `cd ATES_extended; chmod +x run_EXHALE.sh; ./run_EXHALE.sh.`

Original ATES behavior: run as is, with no override files. The result must be identical to upstream ATES.

Activate opacity dispatch:

```
cp inputdata/opacity.inp.example opacity.inp
# edit opacity.inp (e.g. OPACITY_MODEL = T, OPA_FILE_HI = ...)
./run_EXHALE.sh
```

Activate trace metals:

```
cp inputdata/metals.inp.example metals.inp
# edit metals.inp (ion + abundance lines)
./run_EXHALE.sh
# result: output/Metals_ioniz_adv.txt
```

The two side-car files are independent — enable opacity only, metals only, or both at the same time.

Part II

Porting AIOLOS metal cooling into ATES v2.0

(2026-05-28)

8. Overview

This memo documents the changes made to the ATES code (v2.0, branch EXHALE) to incorporate the metal radiative-cooling prescription used in the AIOLOS code (Schulik & Booth 2023, hereafter SB23) and to extend ATES from a pure H/He photoionization-hydrodynamics solver into one that also tracks neutral and singly/doubly ionized carbon and oxygen, with full photo+collisional ionization equilibrium and the associated metal-line cooling.

The work was carried out in 13 incremental stages, each followed by a compile-link verification step. The total touch consists of roughly 700 lines of new or modified Fortran across 14 files in the ATES tree, plus one trailing-backslash fix in the build script.

The implementation is conservative: when both $X_C = 0$ and $X_O = 0$ (the default), the new code paths remain inactive and the executable reproduces the original H/He behavior bit-for-bit.

9. AIOLOS Source Analysis

9.1 Cooling functions

AIOLOS encodes six metal cooling channels as analytic fits in `photochem.cpp:95--132`. In $\text{erg cm}^3 \text{s}^{-1}$ units, per ion times per electron, they read:

$$\Lambda_{\text{CI}}(T) = 10^{-24} + 3.1 \times 10^{-20} e^{-15162/T} [1 + (T/2 \times 10^4)^{1.5}], \quad (1)$$

$$\Lambda_{\text{CII}}(T) = 1.5 \times 10^{-23} + 3.1 \times 10^{-20} e^{-45162/T} [1 + (T/7500)^{1.5}], \quad (2)$$

$$\Lambda_{\text{OI}}(T) = 5.5 \times 10^{-24} + 1.1 \times 10^{-20} e^{-30162/T} [1 + (T/7500)^{0.5}], \quad (3)$$

$$\Lambda_{\text{OII}}(T) = 5.1 \times 10^{-20} e^{-35162/T} [1 + (T/7500)^{0.5}]. \quad (4)$$

The C III, O III, and H_3^+ channels are present as stub functions returning zero.

Tracing through `git log`, all six fits were introduced by a single 2022-03-09 commit (63695ea, “Ionization+Chemistry done”) with no inline citation. The header comment in `photochem.cpp:16` attributes the cooling block to Black (1981), but Black’s paper covers only the H/He primordial plasma, so the metal fits cannot derive from it. The exponential excitation temperatures ($T^* = 15162, 45162, 30162, 35162 \text{ K}$, corresponding to $\Delta E = 1.31, 3.89, 2.60, 3.03 \text{ eV}$) are loosely consistent with the dominant forbidden lines ($[\text{C I}]^3\text{P}^1\text{D}$, $[\text{O II}] 3727 \text{ \AA}$, etc.) but do not match published one-line fits in any reference I could verify in the codebase. Most likely they are analytic approximations to a Cloudy or similar tabulation, derived by the SB23 authors directly. **Provenance verification against the SB23 paper supplement or the Schulik PhD thesis remains pending.**

9.2 Application context and the β escape factor

The cooling functions are applied in `chemistry.cpp:1000--1090` via `c_Sim::do_highenergy_cooling()`, with a rate in each cell

$$\dot{Q}_{\text{cool},X} = n_X n_e \beta \mu \Lambda_X(T_e),$$

where $\mu = \text{photocooling_multiplier}$ (a global tuning parameter, default 1) and β is the line escape probability

$$\beta(\tau) = \begin{cases} \frac{1 - e^{-2.34\tau}}{4.68\tau} & \tau < 7, \\ \frac{1}{4\tau \sqrt{\ln(\tau/\sqrt{\pi})}} & \tau \geq 7. \end{cases}$$

The optical depth is computed as

$$\tau = \kappa_{\text{tot}}(j, 0) \Delta x_j \times 10^8,$$

where κ_{tot} is the band-0 total opacity in cm^{-1} , and Δx_j is the cell width in cm.

9.3 The 10^8 factor: tuning, not unit conversion

A natural first guess is that the trailing factor of 10^8 represents the cm-to- $\text{\AA}$ conversion. Three pieces of evidence rule this out. First, `init_and_bounds.cpp:56--57` comments explicitly state that `PARI_DOMAIN_MIN/MAX` are in cm, and

radiation.cpp:79 uses $\tau = \kappa \Delta x$ without any 10^8 factor in the radiation solver itself. Second, `git log -S` on the line shows the factor was tuned over a one-month period: 10^2 at introduction (commit 63695ea, 2022-03-09), then 10^4 , then the present 10^8 (commit 112bb6c, 2022-04-03). Third, the value 10^8 is consistent with the physical ratio between metal resonance line-center cross sections ($\sim 10^{-13} \text{ cm}^2$) and the band-averaged continuum cross section ($\sim 10^{-21} \text{ cm}^2$); the AIOLOS gray opacity table does not contain metal lines, so this factor compensates.

The ratio coincidentally matches the $\text{cm}/\text{\AA}$ conversion because line widths are of order an angstrom, but the meaning is physical, not unit-conversion. In `util_ion_eq.f90` the AIOLOS form is kept in the source but commented out, tagged with the policy comment from user memory:

```
!To Be Checked/AIOLOS tuning?  
!tau_eff(j) = kappa_loc(j) * dr_cm(j) * 1.0e8  
tau_eff(j) = kappa_loc(j) * dr_cm(j)
```

9.4 Adopted assumption: 100% escape ($\beta = 1$)

With the 10^8 boost active, $\tau_{\text{eff}} \sim 10^6$ in the escape region, so $\beta \rightarrow 0$ and the metal-line cooling is switched off almost entirely — the lines are treated as fully trapped. Because the 10^8 factor is an undocumented empirical boost (see §2.3) and the metal resonance/fine-structure lines are in fact optically thin in the tenuous escaping flow, the EXHALE implementation *overrides* the computed β and adopts the optically-thin limit, assuming 100% photon escape:

```
! AIOLOS multiplies the escape probability by a 1e8 factor  
! (chemistry.cpp:1006), which drives tau_eff up to ~1e6 so that  
! beta_esc -> ~0 and the metal lines are treated as essentially  
! fully trapped (metal-line cooling switched off).  
! Here we instead assume 100% escape (optically-thin limit):  
! beta_esc = 1, so the full metal-line cooling is applied,  
! matching ATES_extended's coronal treatment.  
beta_esc = 1.0d0
```

This makes the MINPACK-coupled metal cooling in EXHALE quantitatively consistent with the decoupled coronal treatment in ATES_extended (which carries no β at all). A β sensitivity scan for HD 209458b (solar C+O) gives $\Delta T_{\text{max}} \approx 0$ with the 10^8 factor, $\approx 1425 \text{ K}$ with the factor removed ($\beta \approx 0.47$), and $\approx 2840 \text{ K}$ with $\beta = 1$; the latter matches ATES_extended's $\approx 2852 \text{ K}$.

10. ATES Architecture Overview

ATES v2.0 solves a one-dimensional, time-dependent photoionization-hydrodynamics problem on a radial grid in $r \in [R_p, r_{\text{max}}]$. The species fraction array `f_sp(1-Ng:N+Ng,k)` carries, in each cell, the mass fraction of each chemical species, originally indexed by $k \in \{1, 2, 3, 4, 5, 6\}$ for {H I, H II, He I, He II, He III, He I(2^3S)}.

The main time loop (`EXHALE_main.f90`) repeatedly calls `ioniz_eq` (in `ionization_equilibrium.f90`) which in turn calls

- PH_heat_HHe (in util_ion_eq.f90): integrates $\sigma_X F_\nu e^{-\tau_\nu}$ over the XUV grid to obtain photoionization rates and the bolometric photoheating.
- eval_cool (in util_ion_eq.f90): assembles the total radiative cooling rate from recombination, collisional ionization, bremsstrahlung, and collisional excitation channels.
- hybrd1 (MINPACK) with one of ion_system_H, ion_system_HeH, or ion_system_HeH_TR: solves the steady-state ionization balance for the N_{eq} -dimensional unknown vector of ion fractions.

The thermal equation (T_equation.f90) is solved separately in post-processing, with cooling rate functions called via scalar *_func variants of the same coefficients.

11. Implementation: Stage-by-Stage

The work was sequenced so that additive-only changes were completed and verified before any signature-breaking modifications.

11.1 Stage 1: Metal photoionization cross sections

File: src/modules/functions/cross_sec.f90.

Added a Verner et al. (1996) single-shell fit

$$\sigma(E) = \sigma_0 F(y) = \sigma_0 [(y-1)^2 + y_w^2] y^{-Q} (1 + \sqrt{y/y_a})^{-P},$$

with $y \equiv E/E_0$ and $Q = 5.5 + l - P/2$, implemented as a single helper sigma_Verner96, plus four thin wrappers sigma_CI, sigma_CII, sigma_OI, sigma_OII. The numerical $(E_{\text{th}}, E_0, \sigma_0, y_a, P, y_w, l)$ tuples are taken from Verner et al. (1996) Table 1 (outer 2p shell) and are flagged **VERIFY** in the source comment.

11.2 Stage 2: Metal recombination and collisional ionization rates

File: src/modules/radiation/Cool_coeff.f90.

Added eight new array subroutines: rec_CII, rec_CIII, rec_OII, rec_OIII (originally Aldrovandi & Pequignot 1973 power-law form, $\alpha(T) = \alpha_4 (T/10^4)^{-n}$; **since replaced by the Badnell RR+DR fits** — see §14.) and ion_coeff_CI/CII/OI/OII (Voronov 1997 fit $Q(T) = A(1 + P\sqrt{U})/(X + U) U^K e^{-U}$ with $U = \Delta E/k_B T$). These supply the ionization-equilibrium rates and (via the appropriate sums) the bremsstrahlung contribution from metal ions.

Cooling fits ported from AIOLOS (Stage 0, completed in an earlier session) consist of four cool_CI/CII/OI/OII subroutines plus their scalar *_func variants for use in T_equation. At $T = 10^4$ K the values reproduce the AIOLOS fits to all displayed digits: $\Lambda_{\text{CI}} = 9.21 \times 10^{-21}$, $\Lambda_{\text{CII}} = 8.76 \times 10^{-22}$, $\Lambda_{\text{OI}} = 1.17 \times 10^{-21}$, $\Lambda_{\text{OII}} = 3.27 \times 10^{-21}$ erg cm³ s⁻¹.

11.3 Stage 3: Global parameters

File: src/modules/init/parameters.f90.

- Two scalar abundance variables: X_C , X_O (number ratios relative to hydrogen; solar 2.7×10^{-4} and 4.9×10^{-4}).
- Boolean toggle `thereis_metals`, set in `input_read.f90` from $X_C > 0$.or. $X_O > 0$.
- Four new allocatable cross-section vectors s_{ci} , s_{cii} , s_{oi} , s_{oii} (sized at runtime to the energy-grid length N_l).

11.4 Stage 4: Energy-grid initialization

File: `src/modules/init/set_energy_vectors.f90`.

In both the power-law SED branch and the monochromatic branch, the metal cross-section vectors are allocated and populated from `sigma_CI/CII/OI/OII(e_v(i))`, immediately after the existing H/He vectors are filled.

11.5 Stage 5: Compile verification of additive-only work

At this point Stages 1–4 introduce only new symbols and new fields; nothing that already existed has been modified in signature or behavior. A full compile-link with 47 object files and ATES executable verifies that the additions are valid Fortran and that the H/He-only behavior is byte-for-byte preserved.

11.6 Stage 6: f_{sp} dimension 6→12

Files: `EXHALE_main.f90`, `src/modules/init/set_IC.f90`, `src/modules/init/init.f90`,
`src/modules/files_IO/load_IC.f90`, `src/modules/radiation/ionization_equilibrium.f90`.

The species array is extended from six to twelve columns by enlarging the second dimension in every declaration and adding the metal indices $k \in \{7, 8, 9, 10, 11, 12\}$ for $\{C\text{ I}, C\text{ II}, C\text{ III}, O\text{ I}, O\text{ II}, O\text{ III}\}$. Initial conditions default the metals to neutral:

$$f_{sp}(:, 7) = X_C / (1 + 4 Y_{\text{He/H}}), \quad f_{sp}(:, 10) = X_O / (1 + 4 Y_{\text{He/H}}),$$

with the higher stages initialized to zero. Loaded IC files (which contain only the original six columns) are similarly padded with neutral metals at load time.

Inside `ioniz_eq` the metal number-density profiles $n_{C\text{ I}}, n_{C\text{ II}}, n_{C\text{ III}}, n_{O\text{ I}}, n_{O\text{ II}}, n_{O\text{ III}}$ are extracted from the new f_{sp} slots and the totals $n_C = n_{C\text{ I}} + n_{C\text{ II}} + n_{C\text{ III}}$, n_O analogously, are formed.

11.7 Stage 7: Photoionization with metal channels

File: `src/modules/radiation/util_ion_eq.f90` (`PH_heat_HHe`) and `src/modules/functions/utilities.f90` (`calc_column_dens_metals`).

A new helper `calc_column_dens_metals` integrates the columns $N_{C\text{ I}}, N_{C\text{ II}}, N_{O\text{ I}}, N_{O\text{ II}}$ inward from the outer boundary. In `PH_heat_HHe` the total optical depth at energy E_i becomes

$$\tau_E = 10^{-18} \sum_X \sigma_X(E_i) N_X,$$

with $X \in \{\text{HI}, \text{He I}, \text{He II}, \text{CI}, \text{C II}, \text{OI}, \text{O II}\}$ (plus the He I triplet term when enabled). Photoionization rates for the four metal species are accumulated analogously to the existing H/He integrals,

$$P_X = 10^{-18} \int F_\nu(E) e^{-\tau_E} \sigma_X(E) \frac{dE}{E},$$

and the bolometric photoheating integrand gains four metal terms

$$\delta \mathcal{H}_X = \sigma_X(E_i) [1 - E_{\text{th},X}/E_i] n_X(j),$$

with $E_{\text{th},X}$ taken from the Verner threshold list.

The subroutine signature is extended; the single caller in `ionization_equilibrium.f90` and the two calls in `post_process_adv.f90` are updated. The post-processing calls pass zero-valued metal density arrays (`zerov`), since post-processing only refines the converged profiles and does not track metal feedback.

11.8 Stage 8: Metal-line cooling with β escape factor

File: `src/modules/radiation/util_ion_eq.f90` (`eval_cool`).

The cooling routine signature now accepts the six metal density arrays and returns the four metal-stage recombination coefficients and the four metal-stage collisional ionization coefficients (for downstream use in the equilibrium system). The bremsstrahlung term is extended with Z^2 -weighted contributions from each ionized metal stage,

$$\dot{Q}_{\text{ff}} = 1.426 \times 10^{-27} \sqrt{T} \left[\sum_Z Z^2 g_Z(T) n_Z \right] n_e, \quad (5)$$

with the sum now including $Z = 1$: {H II, C II, O II} and $Z = 2$: {He II, He III, C III, O III}. The two Gaunt-factor calls `GF(T, 6.0d0, GF_C)` and `GF(T, 8.0d0, GF_O)` reuse the existing $g_Z(T)$ routine.

The metal line-cooling contribution is

$$\dot{Q}_{\text{metal}} = \beta(\tau_{\text{eff}}) n_e [n_{\text{CI}} \Lambda_{\text{CI}} + n_{\text{CII}} \Lambda_{\text{CII}} + n_{\text{OI}} \Lambda_{\text{OI}} + n_{\text{OII}} \Lambda_{\text{OII}}], \quad (6)$$

with the AIOLOS escape probability

$$\tau_{\text{eff}}(j) = \kappa_{\text{loc}}(j) \Delta r_j \times 10^8,$$

$$\kappa_{\text{loc}}(j) = 10^{-18} [\sigma_{\text{HI}}(E_1) n_{\text{HI}}(j) + \sigma_{\text{HeI}}(E_1) n_{\text{HeI}}(j) + \sigma_{\text{HeII}}(E_1) n_{\text{HeII}}(j)].$$

The constant 1.7724 in the high- τ branch of the AIOLOS expression is recognized as $\sqrt{\pi}$ and written that way in Fortran for clarity: `sqrt(log(tau_eff/sqrt(pi)))`. The innermost ghost cell has β forced to zero (matching AIOLOS). The `!To Be Checked/AIOLOS tuning?` comment is placed on the preceding line at the point of the 10^8 factor.

Adopted value. The $\beta(\tau_{\text{eff}})$ expression above is retained in the source for reference, but the production code overrides it with $\beta = 1$ (100% escape, optically-thin limit); see §9.4. Equivalently, Λ_{metal} is applied with no line-trapping suppression.

The total cooling becomes $\dot{Q}_{\text{cool}} = \dot{Q}_{\text{H/He}} + \dot{Q}_{\text{metal}}$.

11.9 Stage 9: New 7-equation ionization system

File (new): src/modules/nonlinear_system_solver/System_HeHCO.f90.

The 7-dimensional unknown vector encodes the ionization fractions

$$x = (x_{\text{HII}}, x_{\text{HeII}}, x_{\text{HeIII}}, x_{\text{CII}}, x_{\text{CIII}}, x_{\text{OII}}, x_{\text{OIII}}),$$

each normalized to the respective element total ($x_{\text{HII}} = n_{\text{HII}}/n_{\text{H}}$, etc.). The neutral fractions follow from conservation: $x_{\text{HI}} = 1 - x_{\text{HII}}$, $x_{\text{HeI}} = 1 - x_{\text{HeII}} - x_{\text{HeIII}}$, $x_{\text{CI}} = 1 - x_{\text{CII}} - x_{\text{CIII}}$, $x_{\text{OI}} = 1 - x_{\text{OII}} - x_{\text{OIII}}$. The free-electron density includes metal ion contributions:

$$n_e = n_{\text{HII}} + n_{\text{HeII}} + 2n_{\text{HeIII}} + n_{\text{CII}} + 2n_{\text{CIII}} + n_{\text{OII}} + 2n_{\text{OIII}}.$$

The seven balance equations follow the same template as the original 3-equation HeH system,

$$f_k = n_{\text{lo}} \Gamma_{\text{lo}} + n_e [n_{\text{lo}} C_{\text{lo}} - n_{\text{up}} \alpha_{\text{up}}] = 0,$$

where Γ is the photoionization rate, C the collisional ionization rate coefficient, and α the case-B recombination coefficient. Defensive code paths force $f_4 = x_4$, $f_5 = x_5$ when $n_{\text{C}} \leq 10^{-30}$ (and analogously for O), keeping the Jacobian well-defined in the limit of absent metals.

The params array dimension is consequently bumped from 25 to 40 throughout all system modules (System_H.f90, System_HeH.f90, System_HeH_TR.f90, System_HeHCO.f90, T_equation.f90, and the caller).

11.10 Stage 10: Equilibrium-system dispatch

File: src/modules/radiation/ionization_equilibrium.f90.

The dispatch logic chooses among three callees:

```
if (thereis_HeITR) then
  call hybrd1(ion_system_HeH_TR, ...)
else if (thereis_metals) then
  call hybrd1(ion_system_HeHCO, ...)
else
  call hybrd1(ion_system_HeH, ...)
endif
```

The HeITR + metals combination is not supported in this revision (a flag check would short-circuit one or the other if both were requested; currently HeITR wins).

The params buffer is filled with the new metal coefficients in slots 19–32: P_X in 19–22, α_X in 23–26, C_X in 27–30, and $n_{\text{C}}, n_{\text{O}}$ in 31–32. Initial guesses for the seven-vector are chosen analogously to the existing pattern (full ionization at the outermost cell, propagation inward).

11.11 Stage 11: Thermal equation

File: src/modules/nonlinear_system_solver/T_equation.f90.

The Newton residual for the temperature equation gains a metal-line cooling contribution (using the scalar cool_*_func forms) and a metal-ion bremsstrahlung sum. The free electron density includes metal contributions when thereis_metals is true. Metal densities are passed via paramST(13--18); the post-processing caller fills these with zero, since the post-process refinement does not propagate metals.

11.12 Stage 12: I/O extensions

Files: files_IO/write_output.f90, files_IO/input_read.f90, files_IO/load_IC.f90.

- Ion_species.txt gains six new columns (C I, C II, C III, O I, O II, O III), appended to the existing six columns. The Hydro_ioniz.txt format is unchanged.
- input_read.f90 sets $X_C = X_O = 0$ at subroutine entry (default), then sets $\text{thereis_metals} = (X_C > 0)$.or. $(X_O > 0)$ after the He/H line is parsed. No new lines are required in input.inp; abundances can be set programmatically or by a future GUI extension.
- load_IC.f90 reads only the original six species columns (preserving compatibility with existing IC files) and initializes the metal fractions from the input abundance, neutral.

11.13 Stage 13: Build and smoke test

File: run_EXHALE.sh.

Compile-link with both gfortran -O2 -fopenmp yields 48 object files (the extra one is System_HeHCO.f90) and a 172520-byte executable, up from 159976 bytes for the unmodified ATES. A smoke run with no input.inp reaches the parameter-reading routine and exits with the expected end-of-file error, confirming module initialization succeeds.

While editing the build script, an unrelated trailing-backslash bug was discovered on line 122 (the line that lists System_implicit_adv_H.f90); without the backslash, the shell variable assignment terminates there and every file below would be silently dropped from the compile string. The backslash was added.

12. Equations in Final Form

12.1 Photoionization integrand

For ion species X the cell-resolved photoionization rate is

$$P_X(j) = 10^{-18} \int_{E_{\text{th},X}}^{E_{\text{top}}} \frac{F_\nu(E) \sigma_X(E)}{E} \frac{e^{-\tau_E(j)}}{1 + a_\tau \tau_E(j)} dE,$$

evaluated by midpoint summation on the N_l -point energy grid. The factor of 10^{-18} restores the cross-section unit, since the tabulated σ_X are stored in units of 10^{-18}cm^2 . The denominator $(1 + a_\tau \tau_E)$ is the ATES bracketing trick for the two-stream approximation; it remains unchanged from upstream.

12.2 Total cooling

$$\begin{aligned} \dot{Q}_{\text{cool}}(j) = & n_e(j) \left[\dot{Q}_{\text{ff}}(j) + \dot{Q}_{\text{coex}}(j) + \dot{Q}_{\text{reco}}(j) + \dot{Q}_{\text{coio}}(j) \right] \\ & + n_e(j) \beta(j) \left[n_{\text{CI}} \Lambda_{\text{CI}}(T) + n_{\text{CII}} \Lambda_{\text{CII}}(T) + n_{\text{OI}} \Lambda_{\text{OI}}(T) + n_{\text{OII}} \Lambda_{\text{OII}}(T) \right]. \end{aligned} \quad (7)$$

The first line contains the legacy H/He channels, now with metal-ion Z^2 terms added to $\dot{Q}_{\text{ff}}$. The second line is the new metal-line cooling; the escape probability is set to $\beta = 1$ (100% escape, §9.4), so this reduces to the full optically-thin metal-line cooling.

12.3 Electron-density closure

$$n_e = n_{\text{HII}} + n_{\text{HeII}} + 2n_{\text{HeIII}} + n_{\text{CII}} + 2n_{\text{CIII}} + n_{\text{OII}} + 2n_{\text{OIII}}.$$

13. Activation, Defaults, and Backward Compatibility

The default configuration is $X_C = X_O = 0$, which yields `thereis_metals = .false.`, which dispatches to the original 3-equation `ion_system_HeH` solver. In this default mode the new code paths execute zero metal contributions to the photoionization, photoheating, opacity, bremsstrahlung, and cooling sums. The original H/He-only ATES behavior is therefore preserved bit-for-bit.

To activate metals, override the defaults in `input_read.f90`:

```
X_C = 2.69d-4 ! solar C/H
X_O = 4.90d-4 ! solar O/H
```

or any other positive value. This single change automatically:

1. Initializes $n_{\text{CI}}, n_{\text{OI}}$ at the lower boundary via `set_IC.f90` and `load_IC.f90`.
2. Switches the equilibrium solver to `ion_system_HeHCO`.
3. Activates the metal photoionization, opacity, photoheating, bremsstrahlung-metal-ion, and metal-line-cooling terms.
4. Adds six metal columns to `Ion_species.txt`.

A future cleanup should propagate X_C/X_O through the `input.inp` text format and the Tk GUI (`src/utis/EXHALE_interface_main.py`); this revision deliberately keeps the input format unchanged for ease of testing.

14. Modernization update (2026): Badnell recombination, nitrogen, two-level cooling

The element-handling physics was subsequently brought in line with the more modern treatment in the sibling fork `ATES_extended`. Three changes were made; all are confined to the metal pathway and leave the H/He-only default

bit-for-bit unchanged.

14.1 Recombination: Aldrovandi–Pequignot → Badnell RR+DR

The Aldrovandi & Pequignot (1973) single power-law fits (radiative recombination only, no dielectronic term) were replaced by the modern **Badnell total-recombination fits**, ported verbatim from ATEs_extended:

- **Radiative** (Badnell 2006, ApJS 167, 334), ground-level total fit

$$\alpha_{\text{RR}}(T) = \frac{A}{\sqrt{T/T_0} (1 + \sqrt{T/T_0})^{1-b} (1 + \sqrt{T/T_1})^{1+b}}, \quad b = B + C e^{-T_2/T}.$$

- **Dielectronic** (Badnell 2006 total fits)

$$\alpha_{\text{DR}}(T) = T^{-3/2} \sum_{i=1}^n c_i e^{-E_i/T}.$$

The fit coefficients are stored as a module-level `rec_fit` derived-type table in `Cool_coeff.f90` (constants `bad_CI`, `bad_CII`, `bad_NI`, `bad_NII`, `bad_OI`, `bad_OII`, keyed by the recombined/daughter ion), evaluated by the helper functions `alpha_rr_metal`, `alpha_dr_metal`, and `alpha_rec_metal`. The existing array routines `rec_CII/rec_CIII/rec_OII/rec_OIII` (and the new nitrogen ones) now loop over cells calling `alpha_rec_metal` with the appropriate daughter label, so the `eval_cool` interface is unchanged. The source fit tables are the Badnell `clist_K` files (see References).

14.2 Nitrogen (N I / N II / N III)

Nitrogen was added end-to-end, mirroring the carbon/oxygen wiring:

- **Photoionization:** Verner et al. (1996) cross sections `sigma_NI`, `sigma_NII` (`cross_sec.f90`), tabulated on the energy grid as `s_ni`, `s_nii`.
- **Recombination:** Badnell RR+DR for the daughters N I and N II (`rec_NII`, `rec_NIII`).
- **Collisional ionization:** Voronov (1997) fits `ion_coeff_NI` ($\Delta E = 14.53$ eV, $A = 4.82 \times 10^{-8}$, $P = 0$, $X = 0.0652$, $K = 0.42$) and `ion_coeff_NII` ($\Delta E = 29.60$ eV, $A = 2.98 \times 10^{-8}$, $P = 0$, $X = 0.310$, $K = 0.30$).
- **Line cooling:** none. Following ATEs_extended (and AIOLOS, which provide no nitrogen fit), `cool_NI` and `cool_NII` return 0; nitrogen participates in the ionization/charge balance and bremsstrahlung but contributes no line cooling. (A future [N II] 122/205 μm treatment via Hollenbach & McKee 1989 is flagged !To Be Checked.)
- **Equilibrium system:** the MINPACK system `System_HeHCO` grows from 7 to **9 unknowns**, adding $x_8 = n_{\text{NII}}/n_{\text{N}}$ and $x_9 = n_{\text{NIII}}/n_{\text{N}}$ (parameter slots 33–39: g_{NI} , g_{NII} , α_{NII} , α_{NIII} , b_{NI} , b_{NII} , n_{N}). Absent elements are force-zeroed, so the same 9-equation system handles C-only, C+O, or C+N+O. $N_{\text{eq}} = 9$ when `thereis_metals`.
- **State / I/O:** the species array `f_sp` grows 12 → 15 (columns 13–15 = N I/N II/N III); `Ion_species.txt` gains three trailing columns (16 total); `set_IC/load_IC` seed neutral N from X_{N} (solar $\simeq 6.76 \times 10^{-5}$); the abundance flag `thereis_metals` now also triggers on X_{N} .

14.3 Two-level fine-structure cooling (optional)

A more physical treatment of the two dominant low-atmosphere coolants, [OI] 63 μm and [CII] 158 μm , was added behind the toggle `use_2lev_cool` (default `.false.`). When enabled, the constant fine-structure part of the Black/AIOLOS fit for O I and C II is replaced by an explicit two-level form with critical-density saturation and neutral-hydrogen collisions,

$$\Lambda = h\nu A \frac{x C_{\text{dex}}}{A + C_{\text{dex}}(1+x)}, \quad x = \frac{g_u}{g_l} e^{-E/k_B T}, \quad C_{\text{dex}} = \sum_c n_c k_{ul}^c,$$

(function `lambda_2level` in `Cool_coeff.f90`); the exponential “forbidden” part of the fit is retained on top, and C I and O II keep the plain fit. This matches `ATES_extended`. **Caveat:** the collision-rate coefficients ([CII]: Goldsmith et al. 2012; [OI]: Lique et al. 2017 / Draine 2011) are approximate and flagged `VERIFY`; the default `.false.` preserves the base fit so that metal-comparison experiments are unaffected unless the toggle is deliberately set.

14.4 Verification

The metal-enabled build (solar C, N, O; HD 209458b) compiles cleanly and runs without NaN/Inf; the nitrogen abundance is conserved to machine precision ($n_N/n_H = 6.76 \times 10^{-5}$) and the N ionization stratification is physical (N I deep, N II through the ionization front, N III in the hot outer wind, tracking hydrogen).

15. Opacity dispatcher, runtime config, and charge exchange (2026)

A second round of work brought across the remaining “modern” pieces of `ATES_extended` and added one process that neither tree had.

15.1 Opacity-model dispatcher (A/C/P/T)

Ported from `ATES_extended` as `src/modules/radiation/opacity_models.f90` (the `photoion_sigma` dispatcher) and `src/modules/files_IO/opacity_input_read.f90` (an `opacity.inp` KEY = VALUE parser). The H/He photoionization cross sections in `set_energy_vectors.f90` are now filled through `photoion_sigma(sp,E)`, which selects:

- **A** — analytic (`cross_sec.f90`; the default, bit-identical to before);
- **C** — constant cross section above threshold, $\sigma = \text{OPA_CONST_X} \times \sigma(E_{\text{th}})$;
- **P** — constant plus Robinson & Catling (2012) pressure broadening (below);
- **T** — tabulated from a two-column `.opa` file per species (E [eV], σ [10^{-18}cm^2]), log–log interpolated, analytic fallback when a species has no file.

Pressure broadening (completed). In `ATES_extended` the pressure factor for each cell was a dormant hook (`opacity_pT_factor` was never called). Here it is actually applied: a multiplier in each cell

$$f(p_j) = 1 + a(p_j/p_{\text{pivot}})^n, \quad p_j = (n_{\text{tot},j} + n_{e,j}) k_B T_j,$$

is stored in the global `opa_pf(j)` (computed in `ioniz_eq / post_process_adv`) and weights the opacity column-density integrand (in `calc_column_dens` and its metals variant), so the optical depth becomes

$$\tau_E(j) = \sum_X \sigma_X(E) \sum_{k \geq j} n_X(k) f(p_k) \Delta r_k.$$

For all models other than **P**, $f \equiv 1$ and the column densities are unchanged (bit-identical).

15.2 Runtime configuration (no recompile)

Two optional files, read next to `input.inp`, remove the need to edit source and rebuild:

- `metals.inp` (`src/modules/files_I0/metals_input_read.f90`) — one ion per line, `<ION> <n_X/n_H>`; `CI/NI/OI` set `X_C/X_N/X_O`. Absent file leaves metals off, so the H/He-only default is preserved. This replaces the previously hard-coded abundances in `input_read.f90`.
- `opacity.inp` — selects the opacity model and its parameters (§15.).

Templates ship in `inputdata/` (`metals.inp.example`, `opacity.inp.example`, `HI_sample.opa`, `README.opacity`).

15.3 Charge transfer with hydrogen (Kingdon & Ferland 1996)

Charge exchange couples metal ionization to the hydrogen ionization fraction and was absent from *both* trees; it is added here to `System_HeHCO.f90` (the coupled MINPACK system is the natural home, since both n_{HI} and n_{HII} are solved simultaneously). Three reaction types per element (C, N, O):



Rate coefficients use the Kingdon & Ferland (1996, ApJS 106, 205) analytic fits (UGA CX database),

$$\alpha(T) = a \times 10^{-9} t_4^b [1 + c e^{d t_4}] e^{-E_k/t_4}, \quad t_4 = T/10^4 \text{ (clamped to the fit range),}$$

with $E_k = E/k$ (in 10^4 K) nonzero only for the endothermic ionization branch (`ct_fit / charge_transfer_rates` in `Cool_coeff.f90`; nine rates passed via `params(40--48)`). Each net rate $\text{cx}_X = \alpha_X^{\text{ion}} n_{X^0} n_{\text{HII}} - \alpha_X^{\text{rec}} n_{X^+} n_{\text{HI}}$ enters the X balance and, with opposite sign, the H balance ($\dot{f}_{\text{HII}} \leftarrow \text{cx}_X$), so the coupling is fully self-consistent. All terms vanish automatically when an element is absent.

Verification. With solar C, N, O the run is stable and NaN-free, and the expected physics appears: oxygen ionization tracks hydrogen tightly ($x_{\text{OII}} \approx x_{\text{HII}}$: e.g. 0.59 vs. 0.57, 0.87 vs. 0.85) because O–H charge transfer is fast and near-resonant, while nitrogen tracks more loosely (its larger energy defect, $E_k = 1.086 \times 10^4$ K, suppresses the ionization branch).

16. Caveats and Verification Items

The following items must be verified before scientific use of the metal-enabled code:

1. **Verner cross-section parameters.** The numerical $(E_{\text{th}}, E_0, \sigma_0, y_a, P, y_w, l)$ values for C I, C II, N I, N II, O I, O II in `cross_sec.f90` should be cross-checked against Verner et al. (1996) Table 1.
2. **Voronov ionization parameters.** The (A, P, X, K) values for the six metal collisional ionization rates (C I, C II, N I, N II, O I, O II) should be cross-checked against Voronov (1997) Table 1.
3. **Recombination fits.** *Resolved:* the Aldrovandi–Pequignot (1973) power-law fits (RR only) have been replaced by the Badnell RR+DR total-recombination fits for all metal stages (see §14.).
4. **AIOLOS β formula and the 10^8 factor.** The AIOLOS form is retained in the source (tagged !To Be Checked/AIOLOS tuning?) but commented out, because the 10^8 factor is an empirical boost rather than a unit conversion (see §2.3). The production code instead adopts $\beta = 1$ (100% escape; see §9.4). Whether the optically-thin assumption is appropriate everywhere in the wind — versus a self-consistent Sobolev/escape-probability treatment — remains to be checked.
5. **Metal cooling fits.** AIOLOS provides no in-code citation; the fits should be located either in SB23 or in the Schulik (2022) PhD thesis before publication.
6. **Post-processing routine.** In `post_process_adv.f90` the calls to `PH_heat_HHe`, `eval_cool`, and `write_output` all pass zero-valued metal density vectors and zero metal output columns. Self-consistency of metals in the post-processing pass is deferred.
7. **He ITR + metals.** The He I triplet pathway and the metal pathway are presently mutually exclusive; He ITR takes precedence in the dispatch.

17. References

- Aldrovandi & Pequignot 1973** A&A 25, 137. Radiative recombination rate coefficients for astrophysical ions. (Superseded here by Badnell; see §14..)
- Badnell 2006** ApJS 167, 334. Radiative recombination data for the isoelectronic sequences (total RR fits); dielectronic data from the adf48 total fits. Tables at <https://amdpp.phys.strath.ac.uk/tamoc/>.
- Goldsmith et al. 2012** ApJS 203, 13. Collisional de-excitation rates for the [C II] 158 μm line.
- Kingdon & Ferland 1996** ApJS 106, 205. Rate coefficients for charge transfer between hydrogen and the first 30 elements.
- Robinson & Catling 2012** ApJ 757, 104. Analytic radiative transfer / pressure-broadened opacity (used for the 'P' model).
- Schulik & Booth 2023** MNRAS 523, 286. AIOLOS: 1-D multi-species radiation-hydrodynamics code description.
- Verner & Ferland 1996** ApJS 103, 467. Atomic data for astrophysics. II. New analytic fits for photoionization cross sections and case-B recombination coefficients.

Verner, Ferland, Korista & Yakovlev 1996 ApJ 465, 487. Analytic single-shell fits for photoionization cross sections of atoms and ions, ground states.

Voronov 1997 ADNDT 65, 1. A practical fit formula for ionization rate coefficients of atoms and ions by electron impact: $Z = 1-28$.

Part III

Trace-metal Phase 2 extensions: next-steps #4–#9 (2026-06-03)

Context note (2026). This memo was originally written for the ATES_extended fork (decoupled coronal metal solver) and has been copied into EXHALE, the main development version. In EXHALE the metal physics lives in the coupled MINPACK ionization system (Badnell RR+DR recombination, nitrogen, charge transfer with H, an A/C/P/T opacity-model dispatcher, and runtime metals.inp/opacity.inp); see cooling_formulas for the closed-form metal line-cooling formulas EXHALE now uses, and Update_EXHALE §7 for the Phase-2 metal line-cooling validation gate. (Updated 2026-08-15: the note originally pointed at ATES_metal_cooling.tex, which no longer exists.)

18. Overview

This memo documents the extensions made to the ATES_extended trace-metal module beyond the initial Phase 2 port, corresponding to next-steps #4–#9 of aiolos_port_memo.pdf. Earlier work (opacity dispatcher, basic 2-stage coronal metals, convergence fix) is covered in aiolos_port_memo.pdf and convergence_fix_and_validation.pdf.

All test runs use HD 209458b (power-law SED, solar He/H, $du_{th} = 2 \times 10^{-2}$) with metals.inp containing OI 4.9e-4.

Summary of changes.

- **#5** Third ionization stage ($X^0/X^+/X^{2+}$); fixed a recombination-daughter bug in the 2-stage solver.
- **#6** Replaced A&P 1973 RR + SVS 1982 DR with Badnell 2006 RR + Badnell DR (exact coefficients from amdpp).
- **#7** Coupled metal-line cooling into the time-loop energy equation (previously post-processing only).
- **#8** Verified the metal cooling effect on $T(r)$.
- **#9** Explicit two-level fine-structure cooling (HI collider + critical-density saturation) for [O I] 63 μm and [C II] 158 μm , combined with the high- T forbidden term.

19. #5 – Three ionization stages and a bug fix

The coronal solver was generalized from two stages (X^0 , X^+) to three (X^0 , X^+ , X^{2+}), restoring element conservation when X^{2+} is non-negligible. The populations follow

$$X^0 : X^+ : X^{2+} = 1 : r_1 : r_1 r_2,$$

$$r_1 = \frac{\Gamma_0}{\alpha_0 n_e}, \quad r_2 = \frac{\Gamma_1}{\alpha_1 n_e},$$

with Γ_0, Γ_1 the photoionization rates of X^0 , X^+ and α_0, α_1 the recombination rates onto X^0 , X^+ . X^{2+} photoionization is neglected (high threshold, small XUV flux), so X^{2+} is a sink. Implemented in `coronal_fractions` (`metals_solve.f90`); X^+ photoionization uses a new `photoion_rate_metal_ion` with the X^+ cross-section grid `sig1` added to `metals.f90`.

Bug fix. The original 2-stage `coronal_ratio` used the wrong recombination coefficient for the $X^+ \rightarrow X^0$ balance: it called `alpha_rec('OII')` (= O III $\rightarrow$ O II) instead of `alpha_rec('OI')` (= O II $\rightarrow$ O I). Because $\alpha(\text{O III} \rightarrow \text{O II})$ is $\sim 6\times$ larger than $\alpha(\text{O II} \rightarrow \text{O I})$, recombination was overestimated and O II underestimated. The fix changes the escape-region ratio from $\text{O I}/\text{O II} \approx 0.94$ (wrong) to ≈ 0.15 (O II dominant), which is physically correct for the strongly-XUV-irradiated escape region.

20. #6 – Badnell recombination rates

Verner & Ferland 1996 covers only H-/He-/Li-/Na-like ions, so it does *not* contain the low-charge C/N/O recombination channels we need. We instead use Badnell's modern total fits, retrieved directly from

https://amdpp.phys.strath.ac.uk/tamoc/{RR,DR}/clist_K

(ground-level $M = 1$ rows). The forms are

$$\alpha_{\text{RR}}(T) = A \left[\sqrt{T/T_0} (1 + \sqrt{T/T_0})^{1-b} (1 + \sqrt{T/T_1})^{1+b} \right]^{-1}, \quad b = B + C e^{-T_2/T},$$

$$\alpha_{\text{DR}}(T) = T^{-3/2} \sum_i c_i e^{-E_i/T}.$$

The `rec_fit` type now stores the six RR parameters plus the variable-length DR (c_i, E_i) arrays. At the escape temperature ($T \sim 10^4$ K) the Badnell rates agree with A&P/SVS to $\sim 10\%$; the improvement is larger at the low temperatures of the deep atmosphere.

21. #7 – Time-loop coupling

Metal-line cooling is now added to the H/He cooling inside `ioniz_eq` (called every time step), immediately after `eval_cool`:

```

if (n_metals > 0) then
  call solve_metals_post(nhi,nhii,nhei,nheii,nheiTR,T_K, nm0,nm1,nm2)
  call eval_metal_cooling(T_K, nm0, nm1, ne, nhi, metal_cool)
  cool = cool + metal_cool
endif

```

so metal cooling enters the time-loop energy update $u_3 \leftarrow u_3 + \Delta t (\text{heat} - \text{cool})$, not only the post-processing pass. A from-scratch HD 209458b run converged in 236,344 iterations with no stiffness problems, confirming the coupling is stable. The path is a no-op when $n_metals = 0$.

22. #8 – Cooling effect on the temperature profile

Comparing metals-off vs metals-on (OI only) with the AIOLOS-fit cooling:

r/R_p	T_{off} (K)	T_{on} (K)	ΔT (K)
1.00 (surface)	1450	1450	0
1.03	1585	1564	21
1.16 (partial ioniz.)	7197	6447	749
2.74 (escape)	3836	3630	206

Metal cooling lowers T , most strongly in the partially-ionized layer ($\Delta T \approx 750$ K), the correct sign and order of magnitude. The value is below the memo’s 1000–3000 K estimate because only O at solar abundance is enabled (no C/N).

23. #9 – Two-level fine-structure cooling

The AIOLOS analytic fit is electron-collider-only and lacks critical-density saturation. For the dominant low-atmosphere coolants [OI] 63 μm (neutral O) and [C II] 158 μm (C^+) we use an explicit optically-thin two-level form:

$$\Lambda = h\nu A \frac{x C}{A + C(1+x)}, \quad x = \frac{g_u}{g_l} e^{-E/k_B T}, \quad C = \sum_c n_c k_{ul}^c,$$

where the collisional de-excitation sum C includes *both* electrons and atomic hydrogen. In the low-density limit $\Lambda \propto C$ (the HI-collider channel), and in the high-density limit $\Lambda \rightarrow h\nu A x / (1+x)$ (critical-density saturation, $n_{\text{cr}} = A/k_{ul}$).

Line data. ([OI] from Draine 2011 / Lique+2017, [C II] from Goldsmith+2012; T-dependence approximate, flagged VERIFY.)

line	A (s^{-1})	E/k_B (K)	g_u/g_l	k_{ul}^e	k_{ul}^H
[OI] 63 μm	8.91×10^{-5}	227.7	0.6	—	$4.2 \times 10^{-11} (T/100)^{0.67}$
[C II] 158 μm	2.29×10^{-6}	91.21	2.0	$8.7 \times 10^{-8} (T/2000)^{-0.37}$	4.0×10^{-11}

Forbidden term. The two-level fine-structure form alone *underestimates* cooling in the hot partially-ionized layer ($T \sim 7000$ K), where collisional excitation of higher levels (e.g. [O I] 6300 Å) dominates. We therefore add the high- T exponential term of the AIOLOS fit (electron collider) on top of the two-level fine-structure term, while dropping the AIOLOS constant fine-structure term (now provided explicitly by the two-level form).

Result. Final cooling model vs the AIOLOS fit (#8) and metals-off:

r/R_p	$T_{2\text{lvl}+\text{forb}}$	T_{fit}	T_{off}
1.07	1946	1905	—
1.16	6380	6447	7196
2.74	3649	3630	—

At the hot layer ($r \sim 1.16$) the forbidden term restores the cooling ($T = 6380 \approx T_{\text{fit}}$, ~ 800 K below metals-off), while in the cooler dense layers ($r \sim 1.07$) the temperature is slightly higher than the fit because critical-density saturation correctly suppresses the over-cooling that the unsaturated fit produced. This is the most physically complete of the three cooling treatments.

24. Caveats and remaining work

1. **Cross-section / rate verification.** The Verner+1996 metal photoionization parameters, the two-level line data (k_{ul}^e , k_{ul}^H , T -dependence), and the dropped-vs-kept split of the AIOLOS fit terms should be checked against primary sources before production.
2. **Forbidden term provenance.** The high- T term reuses the AIOLOS exponential fit, whose origin is itself undocumented (see ATES_metal_cooling.pdf §2). A first-principles multi-level model would be cleaner. [2026-08-15: ATES_metal_cooling.pdf no longer exists; the current account of the AIOLOS C/N/O fits and what replaced them is cooling_formulas §4.]
3. **C/N two-level.** Only O I and C II have the two-level treatment; C I, O II, and N stages still use the AIOLOS fit scaled by n_e . Add their fine-structure lines if those species matter.
4. **X²⁺ cooling.** C III/O III have no cooling fit (consistent with AIOLOS); negligible at these temperatures.

This completes port-memo next-steps #4–#9. The remaining item is #3 (multi-fluid hydrodynamics), deliberately deferred as new physics.